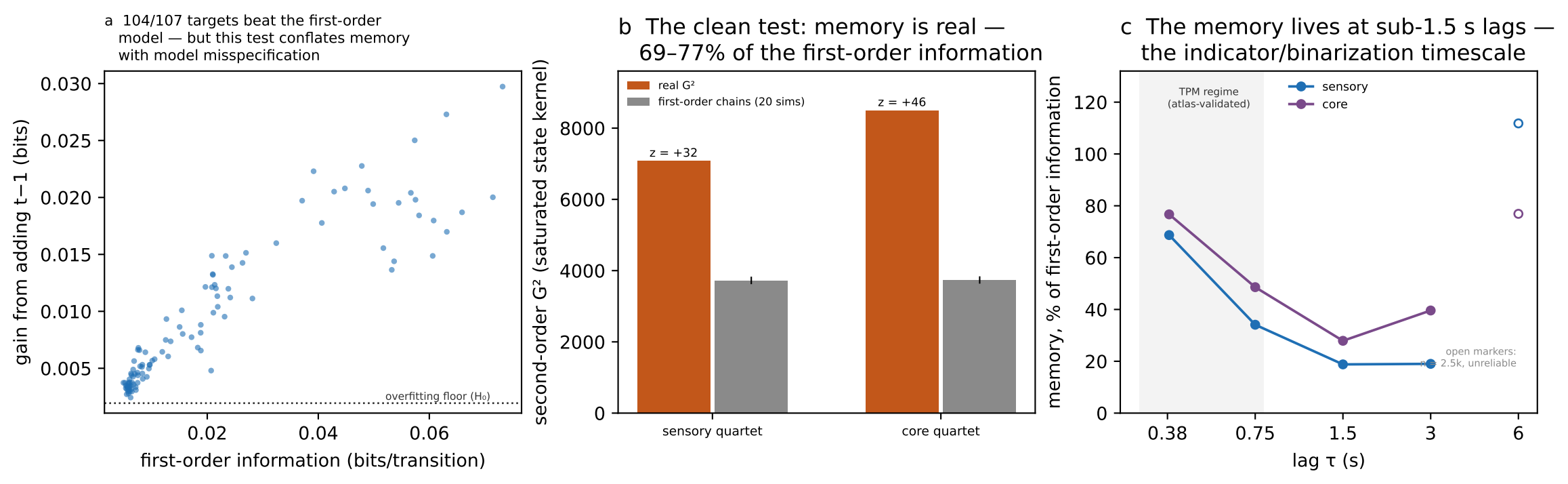
a 104/107 targets beat the first-order model — but this test conflates memory with model misspecification
gain from adding t−1 (bits)
0.030
0.025
0.020
0.015
0.010
0.005
overfitting floor (H₀)
0.02
0.04
0.06
first-order information (bits/transition)
b The clean test: memory is real — 69–77% of the first-order information
second-order G² (saturated state kernel)
real G²
first-order chains (20 sims)
z = +32
z = +46
8000
6000
4000
2000
0
sensory quartet
core quartet
c The memory lives at sub-1.5 s lags — the indicator/binarization timescale
memory, % of first-order information
TPM regime
(atlas-validated)
sensory
core
120
100
80
60
40
20
0
open markers:
2.5k, unreliable
0.38
0.75
1.5
3
6
lag τ (s)